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What if your entire system crashes at 3 AM.. but instead of panic, your system heals itself automatically?

That’s the power of

Site Reliability Engineering (SRE)

⚙ What is SRE?

- SRE (Site Reliability Engineering) is a discipline that uses software engineering to run reliable systems.
- It focuses on automation, monitoring, and scalability.
- The goal is to keep services highly available and stable.
- SRE engineers write code to manage infrastructure instead of manual operations.

🌐 Google's "Special" Connection

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- SRE was invented by Google in 2003.
- Google applied software engineering principles to IT operations.
- Instead of traditional sysadmins, they created engineers who automate reliability.
- Today many companies (Netflix, Uber, Amazon) follow Google's SRE model.

⚔️ DevOps vs SRE

- DevOps → Culture that improves collaboration between development and operations teams.
- SRE → A practical implementation of DevOps principles using engineering practices.
- DevOps focuses on collaboration and CI/CD pipelines.
- SRE focuses on system reliability, monitoring, and automation.

📊 Key Pillars of SRE

(Golden Signals)

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⏱️ Latency

- Measures how fast a system responds to requests.

🌍 Traffic

- Measures how many requests your system receives.

❌ Errors

- Tracks failed requests or system failures.

⚡ Saturation

- Measures how close the system is to its resource limits (CPU, memory, etc.).

📏 Service Level Terms

📌 SLI (Service Level Indicator)

- A metric used to measure system performance (e.g., response time).

🎯 SLO (Service Level Objective)

- The target value for an SLI (e.g., 99.9% uptime).

📜 SLA (Service Level Agreement)

- A formal contract between provider and customer guaranteeing service reliability.

⚖️ Error Budget

- The allowed amount of system failure before reliability targets are violated.

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🤖 Toil & Automation

- Toil = repetitive manual operational work.
- Example: restarting servers, manual monitoring.
- SRE reduces toil through automation and scripts.
- Goal: Engineers focus on building systems, not fixing them repeatedly.

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🚀 Why SRE is Critical in 2026?

- Systems are becoming cloud-native and distributed.
- Applications must scale to millions of users globally.
- Downtime directly causes financial and reputational loss.
- SRE ensures high availability, reliability, and scalability in modern cloud infrastructure

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🚨 Incident Management & Blameless Post-Mortems

- Incident Management handles unexpected system failures quickly.
- Teams follow structured response procedures.
- After incidents, companies conduct Blameless Post-Mortems.
- The goal is learning and improving systems, not blaming individuals.